

Aditya Dutta

Robotics | Embedded Systems |
Rapid Prototyping | Leadership

Skills

Embedded & Electronics Systems

Embedded C/C++, STM32 / ESP32 architectures, low-power firmware design, peripheral interfacing, sensor integration, real-time I/O, hardware debugging.

PCB Design & Hardware Engineering

Multi-layer PCB design (Altium), power system design (Li-ion, charging, regulation), high-density integration, manufacturable layouts, signal integrity considerations.

Robotics & System Integration

End-to-end robotic systems (compute + control + actuation), distributed architectures (Jetson + MCU), motor control (BLDC/servo), feedback systems, sensor fusion.

Mechanical Design & Rapid Prototyping

CAD modeling, mechanism design (bio-inspired + functional), 3D printing, laser-cut fabrication, iterative prototyping under constraints.

Software & Autonomy Stack (beginner)

Python, robotics software structuring, modular architectures, vision-based perception pipelines, hardware abstraction layers.

Control Systems & Mechatronics

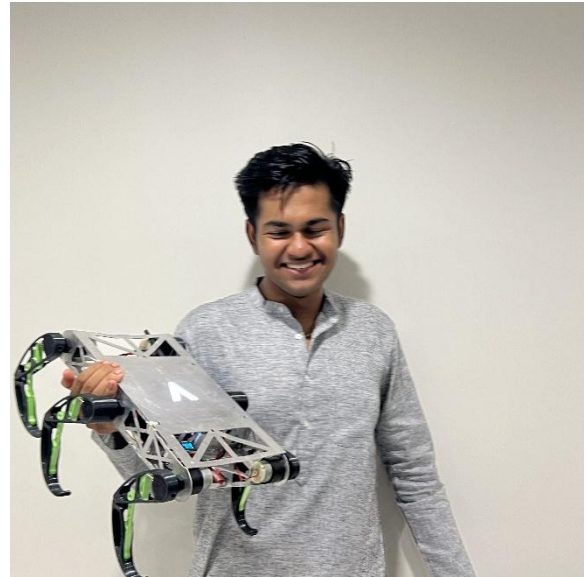
Closed-loop control, encoder feedback systems, motion control, system tuning, real-world constraint handling (latency, noise, power).

Creative Engineering & Prototyping

Zero-to-one system design, constraint-driven innovation, rapid iteration cycles, translating abstract concepts into deployable hardware.

Leadership & Technical Direction

System architecture planning, design reviews, mentoring university engineering teams, breaking down complex systems into executable subsystems.



Information

Undergraduate, Computer Science & Engineering
Manipal Institute of Technology, Bengaluru (MIT-B)
Expected Graduation 2028

Academic Interests

Robotics, Embedded Systems, Autonomous Platforms, Product design and manufacturing.

Current Roles

President — RADAR Student Robotics Club
Co-Founder & Technical Lead — Smart Plant Inc. (Sylvara)

Hometown

Kolkata, West Bengal

Tools and Software Proficiency

Autodesk Fusion 360	Multi-Layer PCB Design
Altium Designer	Advanced CAD
Adobe Photoshop	3D Printing
Oracle SQL DBS	CNC & Laser Manufacturing
BambuSlicer and Cura	Manual PCBA
Matlab (beginner)	Rapid Prototyping
ROS2 Jazzy-Isaac Sim	Product Conceptualization
Linux Terminal	Jetson-based Compute Nodes

CatBot

ONGOING RESEARCH

AUTONOMOUS QUADRUPEL RECONNAISSANCE PLATFORM

OBJECTIVE

Design and develop a lightweight, highly agile autonomous quadruped robotic platform capable of stealthy locomotion, navigation in constrained and unstructured environments, and real-time situational awareness without continuous human supervision.

SYSTEM OVERVIEW

CatBot is a bio-mimetic quadruped inspired by large feline biomechanics, emphasizing superior locomotion, low acoustic output, dynamic balance, and rapid directional control. The platform integrates vision-based perception, inertial sensing, and closed-loop motor control to enable terrain-aware locomotion and autonomous navigation. High-level perception and planning are executed on an onboard NVIDIA Jetson-class compute module, while real-time control and sensor acquisition are delegated to embedded microcontrollers. The system is designed as a compact, self-contained unit with a target gross mass of approximately 2.2 kg.



TECHNICAL SPECIFICATIONS

Processor

NVIDIA Jetson Nano Super (high-level perception)
STM32 / ESP32-class MCUs (real-time control and I/O)

Sensors

Stereo vision module (StereoLabs Zed X Mini),
multi-axis IMU, joint position and multi-node
chassis load sensing

Actuation

BLDC outrunners with planetary gearbox and
modified servo actuators with closed-loop feedback
through magnetic encoders

Framework

Custom robotics software stack
with ultra modular hardware abstraction

KEY ENGINEERING DECISIONS

Vision-Guided Locomotion

Chose vision-first perception over LiDAR to
reduce size, weight, and acoustic signature
while enabling semantic environment
understanding and terrain-aware foot placement.

Bio-Inspired Mechanical Design

A segmented chassis with spine-inspired
compliance to improve balance, shock tolerance,
and agility during dynamic maneuvers.

Modularity and Expansion

Designed a torso-mounted integrated PCB and
modular payload interface to support future
extensions such as extended battery packs,
long-range communication modules, or
cooperative aerial platforms.

Distributed Compute Architecture

Separated high-level autonomy and perception
from real-time motor control to ensure reliable
actuation while maintaining future computational
flexibility.



ONGOING STARTUP : CO-FOUNDER

ULTRA LOW POWER MODULAR PLANT CARE ECOSYSTEM

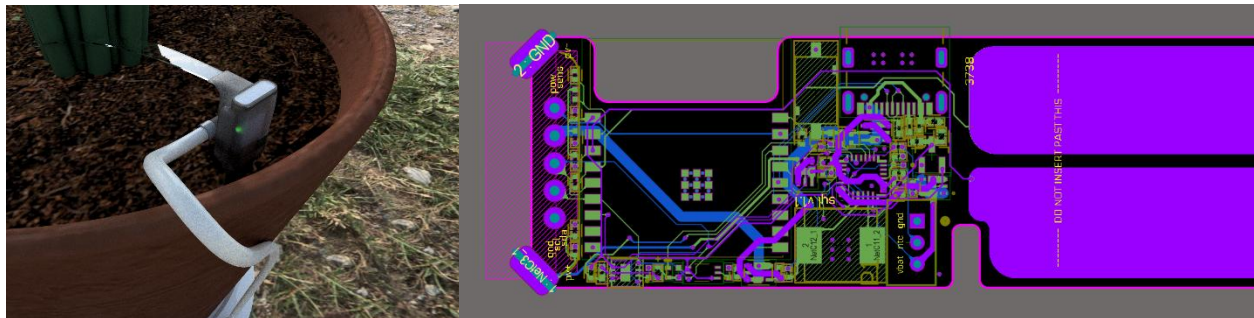
OBJECTIVE

Design and deploy a compact, modular plant-care system capable of year-scale autonomous operation, retrofittable into existing gardening and agricultural setups, while operating under extreme power, size, and cost constraints.

SYSTEM OVERVIEW

[Sylvara](#) is a production-oriented IoT platform focused on long-term, unattended deployment rather than short-lived smart-home demonstrations. The system integrates environmental sensing, soil monitoring, energy harvesting, and wireless connectivity into a highly compact form factor. Designed as a modular ecosystem based around the head unit which is currently in beta-testing, Sylvara prioritizes ultra-low power consumption, field reliability, and manufacturability.

The startup, incorporated in Canada initially as **Smart Plant Inc.**, received **CAD 10,000 in funding** in the summer of 2025, is now trademarked and registered (BCA - Articles of Incorporation - SYLVARA INC. - OCN:1001215100 - April 23, 2025), and has active venture interest offering **CAD 50,000 for 5% equity**, with the support of Morrisette Institute for Entrepreneurship.



TECHNICAL SPECIFICATIONS

Microcontroller

ESP32-C3-WROOM-02 (Wi-Fi, BLE, RISC-V)

Power System

TI BQ25798 based Li-ion battery, solar energy harvesting, USB-C PD charging. Ultra-low-power with multi-month to ~1-year battery life in zero sunlight, unlimited operation with at least 20 minutes of sunlight per day.

Sensing

Temperature, Humidity, Light, Soil Moisture, Capacitive water level monitoring

PCB Electronics Footprint

Approximately 22–35 mm, multi-layer integrated design

Connectivity

Wi-Fi / BLE with cloud backend integration

Sylvara™

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KEY ENGINEERING DECISIONS

Ultra-Low-Power System Design

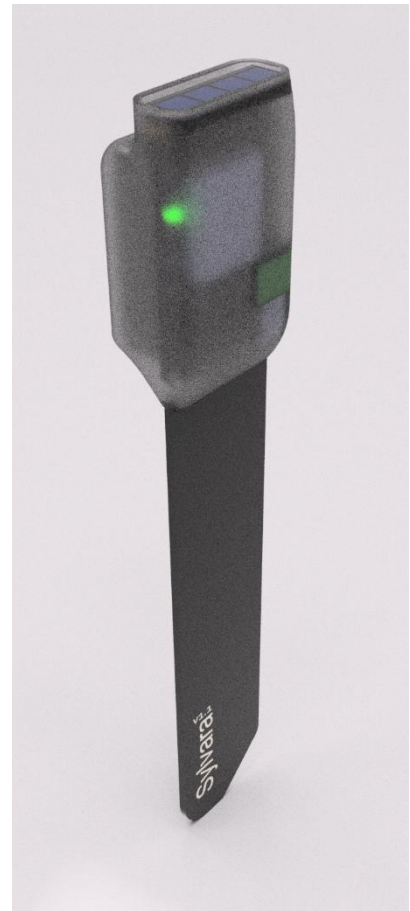
Architected the firmware and hardware around aggressive sleep cycles, peripheral power gating, and duty-cycled sensing to enable near-perpetual operation with ~20 minutes of daily sunlight.

High-Density Integration

Consolidated sensing, power management, charging, and compute onto a single compact PCB using carefully selected ICs to minimize footprint while preserving reliability and serviceability.

Energy-First Architecture

Prioritized power budgeting and energy flow as first-class design constraints, shaping component selection, firmware scheduling, and communication strategy.



Production-Oriented Engineering

Designed the system with manufacturability, enclosure integration, and iterative scaling in mind, bridging the gap between prototype electronics and deployable commercial hardware.

Currently, first batch of 100 being sent in for production.

RoVolt

COMPLETED PROJECT

WHEG-BASED ROBOT AND MOBILE AGRICULTURAL PLATFORM

OBJECTIVE

Design and build a terrain-adaptive robotic platform capable of reliable locomotion across uneven farmland, later extended into a smart agricultural system supporting soil interaction, sensing, and basic farming operations.

SYSTEM OVERVIEW

Revolt originated as a Class 11 robotics project focused on robust mobility using a WHEG (Wheel-Leg Hybrid) locomotion system. The platform was fully CAD-modeled and fabricated using laser-cut aluminum components, manually brazed to form a rigid yet serviceable chassis. Locomotion was achieved using custom 3D-printed WHEGs with a push-rod suspension mechanism, driven by DC motors with encoder feedback and controlled by a Raspberry Pi-based system.

The platform demonstrated stable traversal across varied terrain, including loose soil and uneven surfaces. For public demonstrations and competitions, the system was adapted into a smart agricultural unit by reconfiguring the original 2x3 WHEG layout into a three-wheel configuration and integrating a rear-mounted robotic arm equipped with soil moisture sensing, ploughing, and seeding attachments. Directional awareness was enhanced using an IMU, while existing encoders were reused for motion feedback. A companion mobile application and backend system enabled real-time monitoring, with hardware integration and low-level firmware support led by me.

DEMONSTRATED IMPACT

The Revolt platform was successfully demonstrated at multiple expos and competitive hackathons, achieving top placements and awards. Live evaluations highlighted its reliable terrain traversal, modular reconfiguration, and functional agricultural tooling, validating the system's robustness and design versatility.



RoVolt

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TECHNICAL SPECIFICATIONS

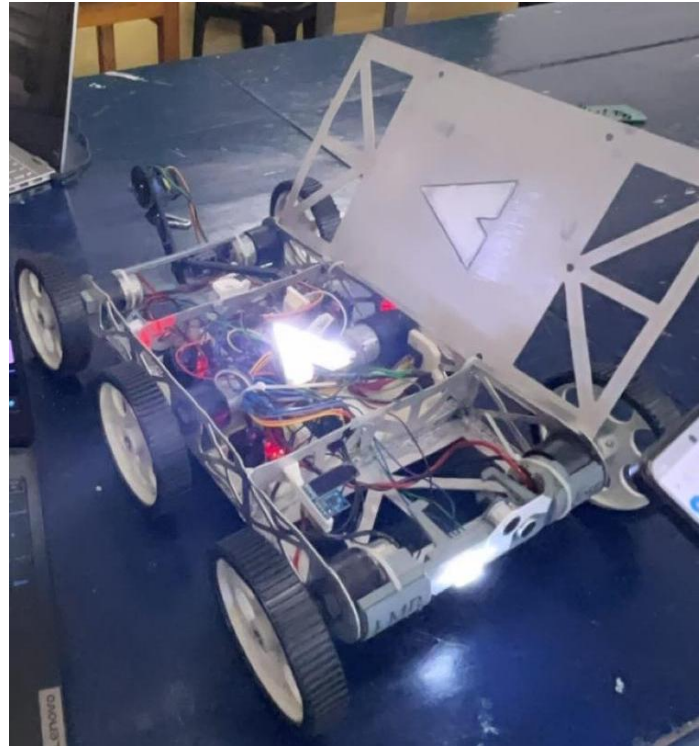
Microprocessor: Raspberry Pi 3 with proto-board.

Control: TB6612FNG dual motor drivers with quadrature encoders.

Locomotion: Custom 3D-printed WHEGs with push-rod suspension.

Sensors: Wheel encoders, IMU, soil moisture sensor, Arm kinematics.

Fabrication: Laser-cut aluminum chassis, manually brazed assembly, 3D-Printed internal mounts and WHEGs



KEY ENGINEERING DECISIONS

WHEG-Based Locomotion

Selected a wheel-leg hybrid design to balance mechanical simplicity with superior terrain adaptability compared to conventional wheeled platforms.

Hands-On Fabrication

Chassis was designed, laser-cut, and brazed manually to ensure structural integrity and rapid iteration without reliance on prefabricated frames.

Platform Adaptability

Reconfigured the locomotion layout and added a modular agricultural arm to demonstrate functional versatility beyond mobility alone.

Sensor Reuse and Expansion

Leveraged existing encoders while integrating inertial sensing to enhance directional stability and reduce system complexity.

Mentorship & Leadership

PRESIDENT MITB's STUDENT ROBOTICS CLUB

As President of the Robotics Club (RADAR), one of the most active technical clubs in Manipal Institute of Technology, Bengaluru. I mentor and guide multiple student teams, translating ambitious ideas into technically feasible systems while accounting for skill-level and resource constraints. Ongoing mentored projects include:

- **TrackMate** — an autonomous athletic track pacer robot focused on consistent velocity control and lap accuracy, Collaboration with Tenvik Sports and in talks with VCs.
- **Arachne** — a spider-robot reconnaissance platform inspired by bio-mimetic locomotion and terrain adaptability, a scaled down mirror to CatBot.
- **BattleRobot Team** — competitive combat robotics involving high-current power systems, mechanical durability, and failure-resilient design.

My role emphasizes architectural guidance, design reviews, subsystem decomposition, and helping teams bridge the gap between concept and execution.

COMPLETED CO-DESIGNED AND MENTORED PROJECTS

DoorThing — Retrofit Electronic Door Unlock System

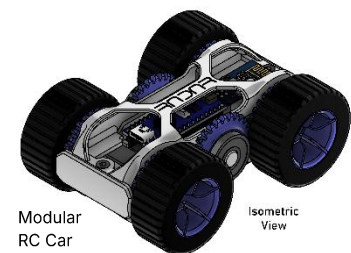
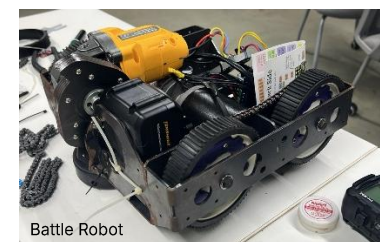
Designed a low-cost, ID-card-based door unlock retrofit system focused on reliability, power efficiency, and ease of installation. The project emphasized embedded control, secure actuation, and practical deployment constraints in institutional environments.

RC Vehicle Platform — Event Robotics Kit

Developed an iterable modular RC vehicle platform for a student robotics event, focusing on robustness and modularity, enough to be repurposed in a pinch for the BattleRobot, is based on beginner-friendly Arduino and NRF modules and integration through a compact interconnection PCB. The platform served as an entry point for teaching drivetrain fundamentals, motor control, and basic embedded interfacing.

Outside engineering, I maintain long distance endurance training and regular exercise as a means of sustaining discipline, recovery, and long-term cognitive performance.

Currently on the track team, placed 3rd in a recent 21k marathon.



Conclusion

ENGINEERING TRAJECTORY AND INTENT

My work in robotics follows a clear lineage rather than a collection of independent projects. **Rovolt** was the origin point: a WHEG-based mobile robot that demonstrated reliable locomotion across uneven terrain and was later adapted into a functional agricultural platform with sensing, actuation, and field interaction. Building Rovolt end-to-end—mechanically fabricated, electrically integrated, and iteratively modified—established my grounding in how robots physically engage with real environments. Its success across multiple expos and hackathons validated both the platform’s adaptability and my systems-level approach to problem solving.

From Rovolt emerged two deliberate and divergent evolutions. **Sylvara** represents the agricultural branch, shifting the core challenge from mobility to endurance, integration, and scalability within a funded startup context. It required energy-first thinking, ultra-low-power electronics, dense PCB integration, and production-oriented design. In contrast, **CatBot** represents the autonomy-driven branch—pushing toward agile locomotion, perception-guided decision-making, and real-time control under strict size, weight, and reliability constraints.

Together, these projects reflect a progression in both technical scope and engineering responsibility: from a mobile robotic platform, to a commercial-grade embedded system, to a research-driven autonomous robot. This trajectory has shaped my engineering philosophy—prioritizing deployable systems, constraint-aware design, and holistic integration over isolated technical novelty.

Moving forward, my goal is to continue developing autonomous robotic systems that extend human capability in environments where risk, scale, or endurance limit direct human presence.
